Assignment 10: Automated Database Backup and Disaster Recovery

Objective: Implement a backup and recovery strategy for MySQL databases using automated backups, retention policies, and recovery procedures.

GitHub Repository Structure

assignment10-backup-recovery/
■■■ backup.sh
■■■ cron/
■ ■■■ backup-cron.txt
■■■ backups/
■■■ docs/
■ ■■■ recovery-procedure.md
■ ■■■ retention-policy.md
■ ■■■ troubleshooting-guide.md
■■■ docker-compose.yml
■■■ screenshots/
■■■ README.md

Implementation Overview

A MySQL backup and disaster recovery solution was implemented using Docker. The solution performs automated database backups, compresses backup files, applies retention policies, and provides documented recovery procedures.

Tasks Completed

✓ Created automated backup script
✓ Scheduled backups using Cron
✓ Compressed backup files
✓ Implemented backup retention policy
✓ Documented recovery procedure
✓ Demonstrated successful backup and restore process

Backup Script

mysqldump -uroot -proot123 companydb > /backup/companydb_backup.sql gzip companydb_backup.sql find /backup -name "*.gz" -mtime +7 -delete

Cron Configuration

0 2 * * * /opt/scripts/backup.sh >> /var/log/mysql-backup.log 2>&1 This configuration executes backups daily at 2:00 AM.

Backup Retention Policy

• Daily backups retained for 7 days.
• Older compressed backups automatically deleted.
• Storage usage controlled through retention automation.

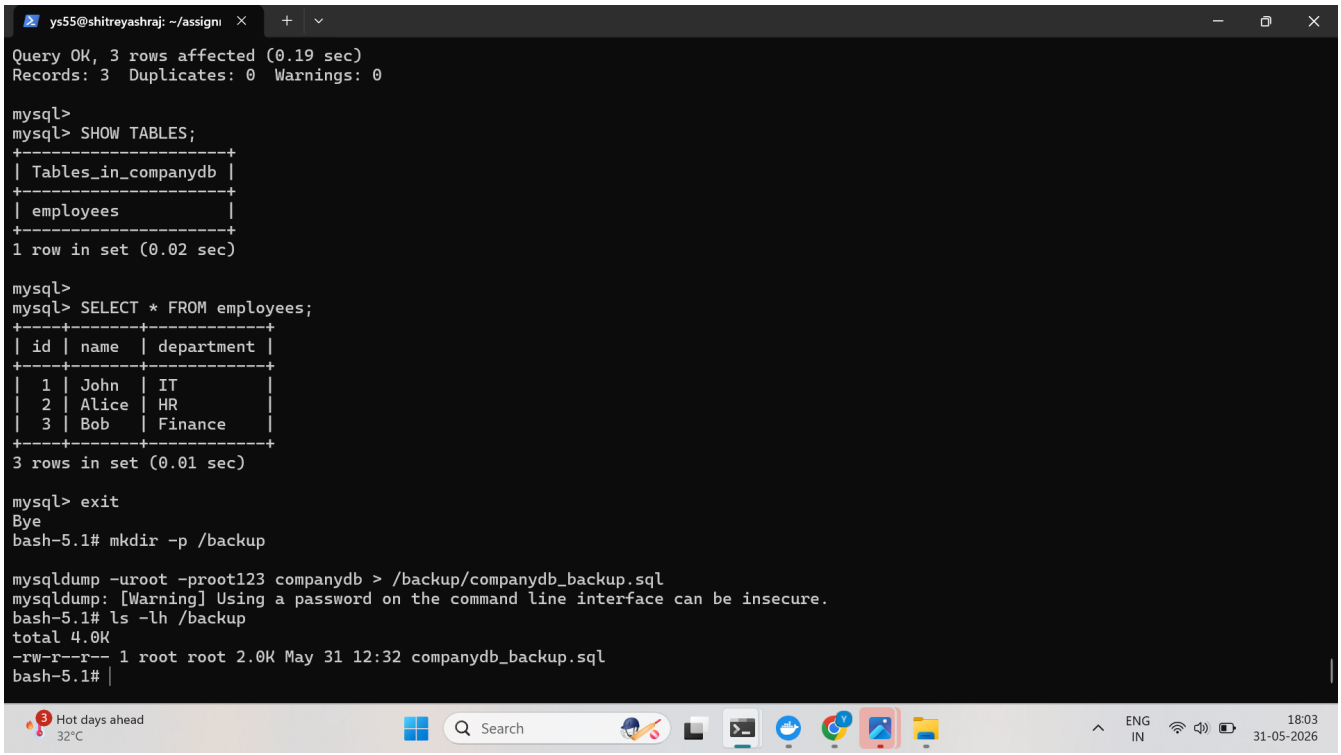
Recovery Procedure

1. Create or restore target database.
2. Locate latest backup file.
3. Restore using MySQL import command.
4. Verify tables and records.
5. Confirm application functionality.

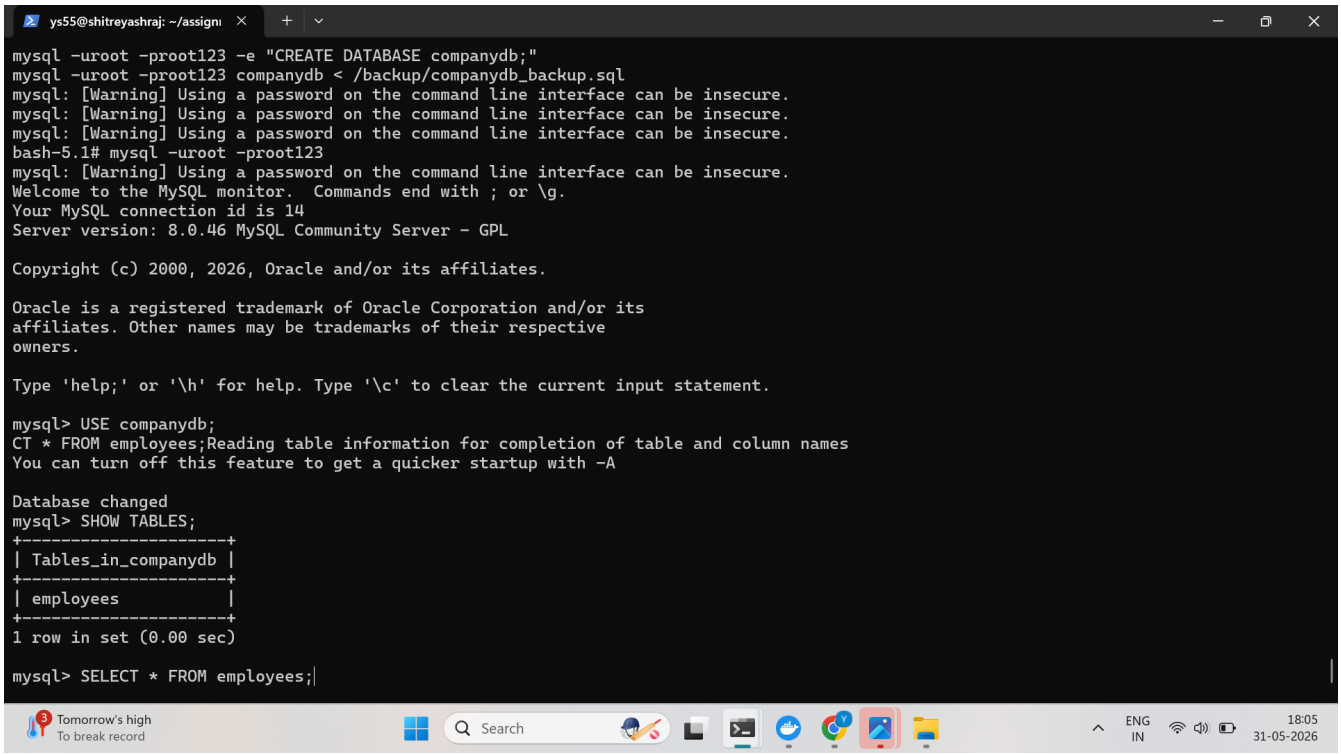
Recovery Commands

mysql -uroot -proot123 -e "CREATE DATABASE companydb;" mysql -uroot -proot123 companydb < companydb_backup.sql

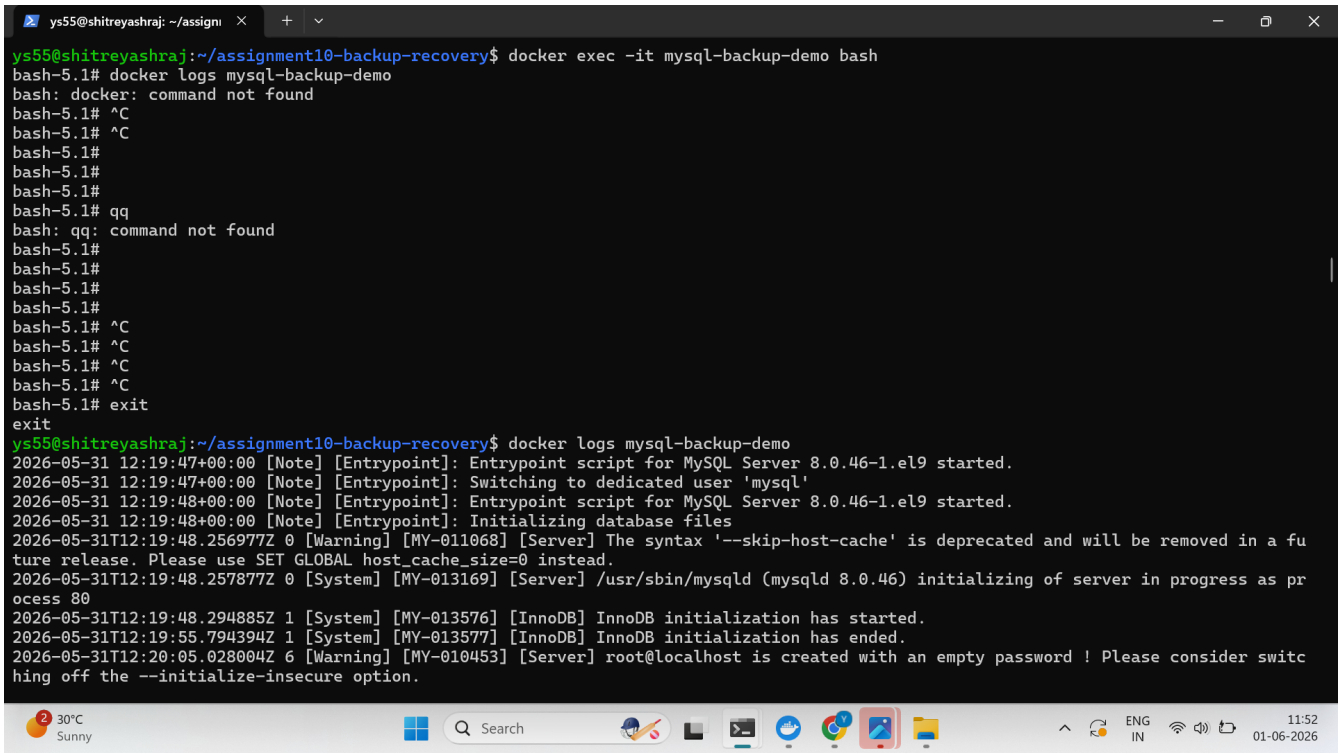
Implementation Evidence



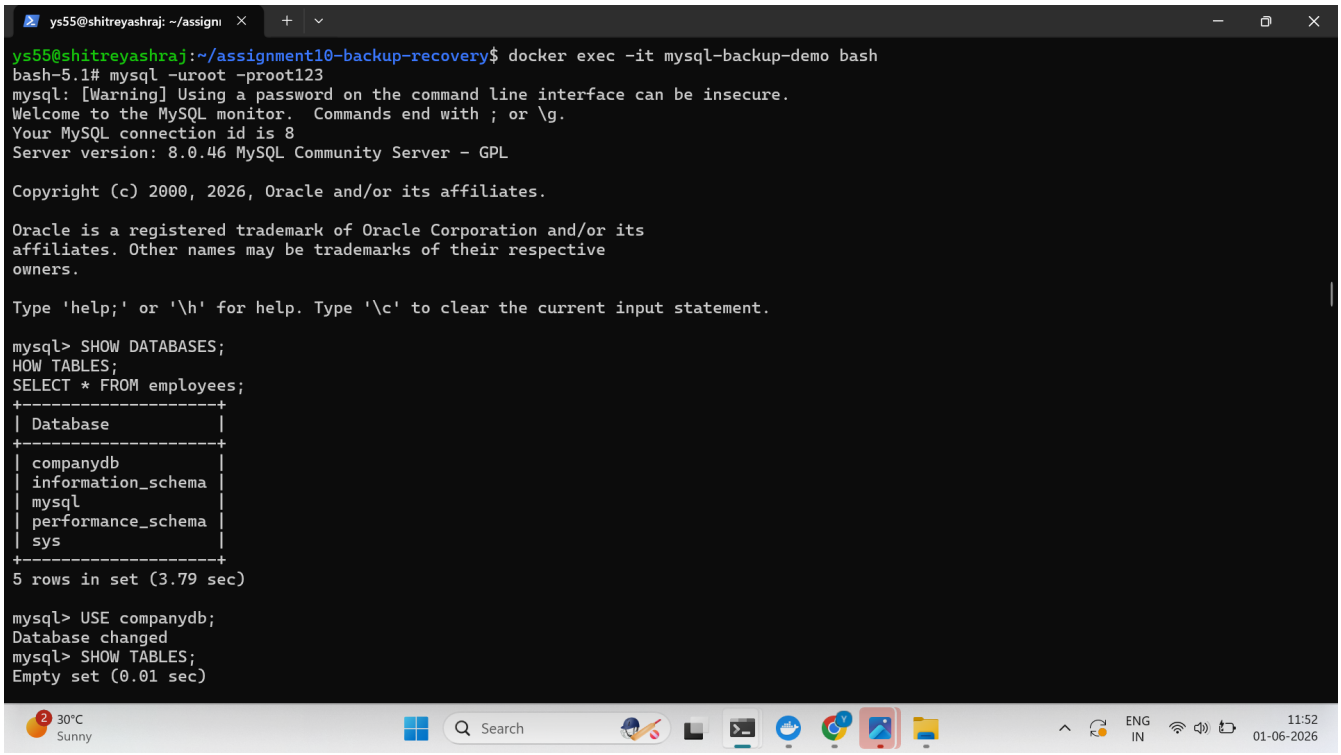
Implementation Evidence



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Testing Results

✓ MySQL container deployed successfully.
✓ Sample database and employee records created.
✓ Backup file generated successfully.
✓ Backup file stored in backup directory.
✓ Database restoration completed successfully.
✓ Employee records verified after recovery.
✓ Disaster recovery process validated.

Recovery Demonstration Summary

The companydb database containing employee information was backed up using mysqldump. The backup file was stored successfully and later restored into MySQL. After recovery, the employees table and all records were verified, demonstrating successful disaster recovery capability.